

1. Scraping Strategy

The pipeline gathers structured content from three different sources: PubMed articles, YouTube videos, and blogs. Each of these sources calls for a different extraction strategy.

1.1 Scraping blogs

To get around bot-detection blocks, blog pages are retrieved using the requests library with a User-Agent header similar to that of a browser. Metadata extraction is done via a prioritised fallback chain:

- A. Main: Before removing any script tags, structured data embedded in page scripts is parsed because LD+JSON (application/ld+json) consistently provides author and datePublished fields in schema.org format.
- B. Fallback: If LD+JSON is absent or insufficient, the scraper searches for standard <meta> tags (name="author", property="article:published_time"), followed by CSS class patterns like byline, entry-date, and post-author, and finally HTML5 <time> elements.
- C. Content extraction targets semantic containers (<article>, <main>, role="main", .post-content, .entry-content) and collects <p> elements longer than 30 characters, removing navigation, footers, and advertisements.

1.2 Scraping YouTube

Open Graph and itemprop meta tags in the page HTML are used to parse YouTube video metadata, including title, channel name, description, and publication date.

YouTubeTranscriptApi is used to retrieve video transcripts, which are then combined into a single text string. The video description serves as a backup content source for downstream tagging and chunking when there is no transcript available (for example, when transcription is turned off).

1.3 Scraping PubMed

Instead of directly scraping the PubMed website, the scraper makes use of the NCBI eUtils API, a free, authenticated-free REST API. The article title, complete author list, journal name, abstract, and publication year are all contained in structured XML that is retrieved by calling the efetch endpoint after the PMID is extracted from the URL using a regular expression. To find out how many times a paper has been cited, you can use the elink endpoint. Specifically, you can look for the pubmed_pubmed_citedin link set. This will give you a list of PubMed articles that cite the paper you're interested in.

2. Topic Tagging Method

Topic tags are created using a LangChain call to GPT-3.5-turbo (utils/tagging.py). The first 500 characters of the content are sent along with the article title in a prompt that asks the model to return 3 to 6 relevant keywords as a raw JSON array. This method is intentionally simple; sending only a short snippet keeps API costs low while giving the model enough context for effective tagging. The model receives instructions via a system message to act as a topic tagger and return only a JSON array, avoiding any preamble or explanation. This makes the output easy to use. If the LLM call fails for any reason, say, a network issue, a missing API key, or a

response that doesn't quite fit, the function will simply return an empty list. This design choice ensures the rest of the pipeline continues to function without interruption. For instance, a healthcare blog might see output like this: ["ketogenic diet", "nutrition", "weight loss", "metabolism", "health"].

3. Trust Score Algorithm

The trust scoring module (scoring/trust_score.py) uses a normalised scoring system between 0.0 and 1.0 for the sources. The scoring algorithm is customised based on the type of source to match the signals of credibility. Another thing that I added on my own was a semantic consistency check. This is the idea that a trustworthy source should be one where the content is actually similar to the title. To do this, I embedded the title and content chunks using OpenAI Embeddings and compared their similarity using cosine similarity. If they are not similar, the trust score is penalised, as this is a common sign of clickbait content

3.1 Scoring Signals

Domain Authority: A three-tier list of credible domains is created. Tier 1 includes harvard.edu, nature.com, and nih.gov. These domains are scored at 0.95. Tier 2 includes healthline.com, bbc.com, etc. These domains are scored at 0.75. Tier 3 includes medium.com and youtube.com. These domains are scored at 0.55. Unknown domains are scored at 0.35. A slight bonus is provided to .edu and .org domains.

Author Credibility: A list of credible organisations is created. A high score of 0.9 is provided to PubMed authors, as they are credible. A score of 0.2 is provided to unknown authors.

Channel Credibility (for YouTube only): Here, the domain is the same for all YouTube videos. So, channel credibility is looked up. A table is created for the same.

Recency Score: The publication year is extracted from the date string and then compared against the current year. If the publication was in the last year, it scores 1.0, gradually decreasing down to 0.2 for publications more than 10 years ago.

Citation Score (Only for PubMed): This score is determined by the number of citations a publication has received. Publications with over 100 citations receive a score of 1.0, and it decreases to 0.2 for publications with fewer or no citations.

Medical Disclaimer Score: Phrases such as "not medical advice," "consult a doctor," and "for informational purposes only" are searched in the publication content. If any of these phrases are found, the publication is given a score of 1.0. A penalty of 0.5 is imposed in case these phrases are not found.

Semantic Consistency Penalty: The average cosine similarity of the title embedding and up to five content chunk embeddings is calculated using OpenAI Embeddings. If the similarity is low, indicating clickbait or non-relevant content, a 30% reduction is applied to the raw score if similarity < 0.3, and a 15% reduction if similarity is between 0.3 and 0.5.

3.2 Weighted Formulas by Source Type

Blog: $0.30 \times \text{author} + 0.30 \times \text{domain} + 0.25 \times \text{recency} + 0.15 \times \text{disclaimer}$

YouTube: $0.50 \times \text{channel credibility} + 0.30 \times \text{recency} + 0.20 \times \text{disclaimer}$

PubMed: $0.25 \times \text{author} + 0.20 \times \text{domain} + 0.15 \times \text{recency} + 0.25 \times \text{citations} + 0.15 \times \text{disclaimer}$

4. Edge Case Handling

4.1 Missing Metadata

- A. Missing Author: 0.2 credibility score is assigned. The pipeline proceeds. The "unknown" value is assigned to the field.
- B. Missing Publish Date: Default 0.3 recency score is assigned. The regex pattern safely matches no year.
- C. Missing Transcript: Fallback to the video's description for language detection, tagging, and chunking.
- D. Empty Content: No errors are raised by `chunk_text()` and `get_topic_tags()`.

4.2 Multiple Authors

PubMed articles often have many authors. In the XML data, the authors are captured. A comma-separated list of authors is created. The credibility score is based on the list of authors. However, the logic is equivalent to the group of authors being a single entity under peer review (all have the same 0.9 credibility score).

4.3 Non-English Content

Automatic language detection is done using the `langdetect` library on the text of the content. The language is detected before the rest of the pipeline. The identified language is recorded and saved under the "language" field. The remainder of the pipeline processes the text, regardless of the identified language. GPT-3.5-turbo supports multilingual text.

4.4 Long Articles

The content is split into chunks using LangChain's "RecursiveCharacterTextSplitter" class, where the chunk size is 1500 characters and the overlap is 150 characters. The text splitter adheres to text boundaries in the following order of priority: paragraph breaks "\n\n", line breaks "\n", sentence endings ".", ",", then words, and finally hard cuts in characters.

4.5 Abuse Prevention

There are several mechanisms included to prevent the system from being exploited by abusers:

- A. Fake or Unknown Authors: If the authors are not included in the list of known credible authors, the credibility score is reduced (0.5 for unknown names, 0.65 for unknown full names, 0.2 for unknown authors). Only sources from trusted organisations are awarded a high credibility score.
- B. SEO Spam Blogs: A default of 0.35 is assigned to the domain authority of unknown domains not included in the tier lists. New domains are penalised without the need for a block list.
- C. Misleading Medical Content: Health-related sources without a medical disclaimer are awarded a 0.5 disclaimer score instead of the maximum 1.0. A downward force is exerted on the final trust score.
- D. Outdated Information: The recency scoring curve penalizes sources heavily for old information. Articles older than 10 years are awarded only 0.2 for the recency score. The overall trust score is reduced by the strong penalty.

- E. Clickbait Titles: Semantic consistency scoring penalizes sources whose titles are weakly correlated with the embedded content (possible cases of clickbait, baits, etc.). Up to 30% of the multiplier is reduced from the raw score.